



12.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Rewriting center-radius form to standard form	  
Key features of a circle	  
Completing the square	  
Rewriting standard form to center-radius form	  
Set-builder notation	  
Interval notation	  

Unit 12, Lesson 9: Graphing Circles



Benchmark

MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.



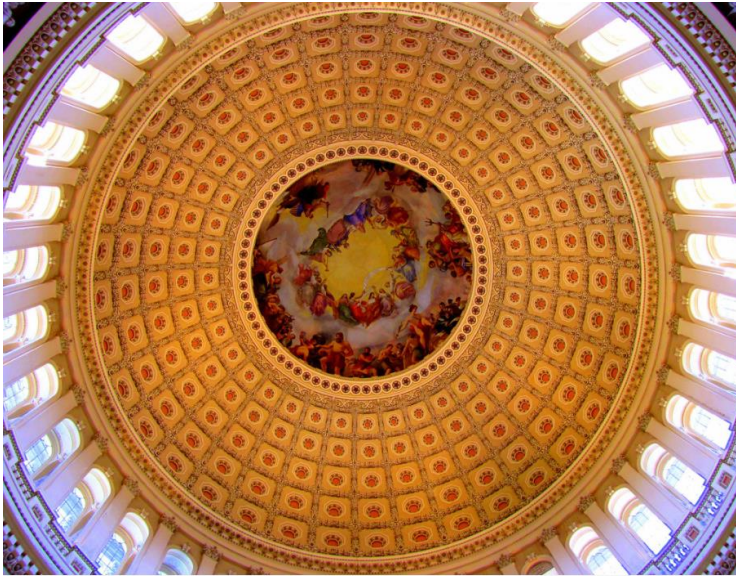
Learning Targets

- ☐ I can graph a circle and determine its key features when given an equation of a circle.
- ☐ I can solve mathematical and real-world problems that are modeled with an equation of a circle.



12.9.1 Warm-Up: Notice and Wonder

The dome of the U.S. Capitol Building is 288 feet (ft) in height and 96 ft in diameter. The dome covers the Rotunda, a room with a circular plan. The architect created an orchestration of architecture, sculpture, and painting in the U.S. Capitol Rotunda.



1. What do you notice?
2. What do you wonder?



12.9.2 Guided Practice: Graphing Circles

1. The equation of circle W is shown.

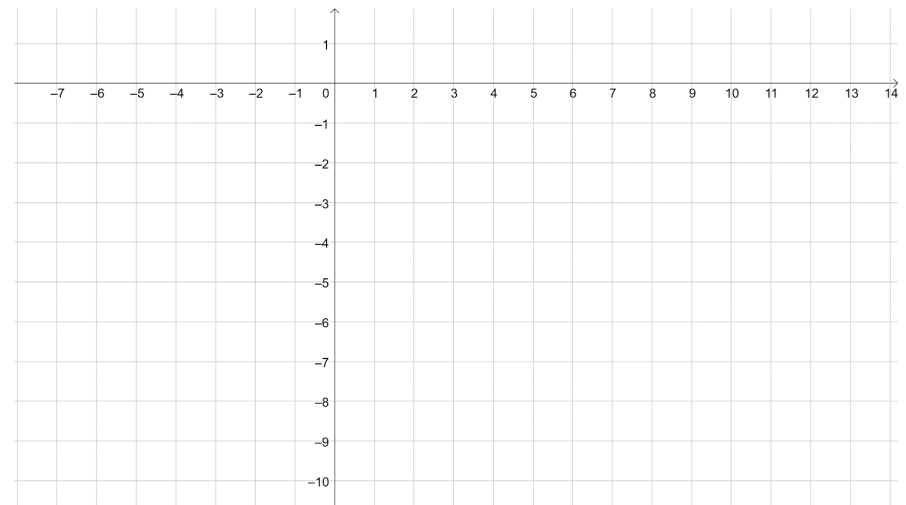
$$x^2 + y^2 - 4x + 10y + 13 = 0$$

- a. Write the equation in center-radius form.

- b. Complete the table.

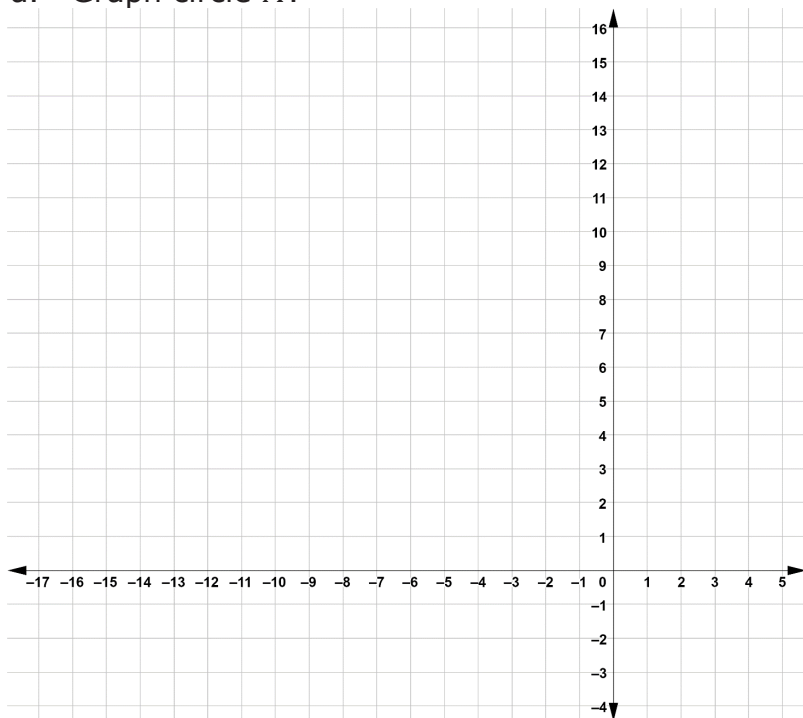
Center	Radius

- c. Graph the circle.



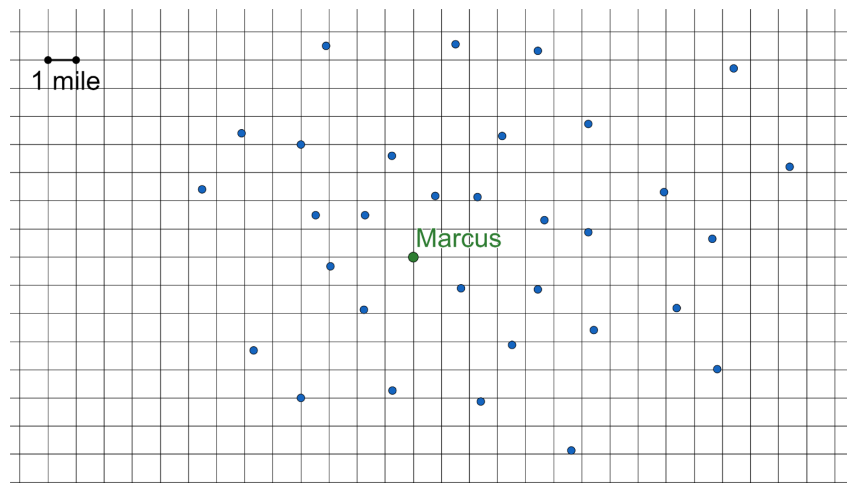
- d. Find the domain of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.
- e. Find the range of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.
2. Marcus wanted to order from a food delivery service and used the equation $(x + 4)^2 + (y - 7)^2 = 64$ of circle M to find food within his radius, in miles.

a. Graph circle M .



- b. What does the center represent in the context?
- c. What does the radius represent in the context?
- d. If the service charges \$3.25 for delivery for every four miles driven, what is the most that Marcus could be charged for the delivery?
- e. Determine the domain of $(x + 4)^2 + (y - 7)^2 = 64$. Write the domain in set-builder notation.
- f. Determine the range of $(x + 4)^2 + (y - 7)^2 = 64$. Write the range in set-builder notation.

- g. The grid shows Marcus' location in green. The blue dots are the locations of restaurants that deliver using UberEats. Use the grid to demonstrate how Marcus could use the domain and range to determine which restaurants he can order from.



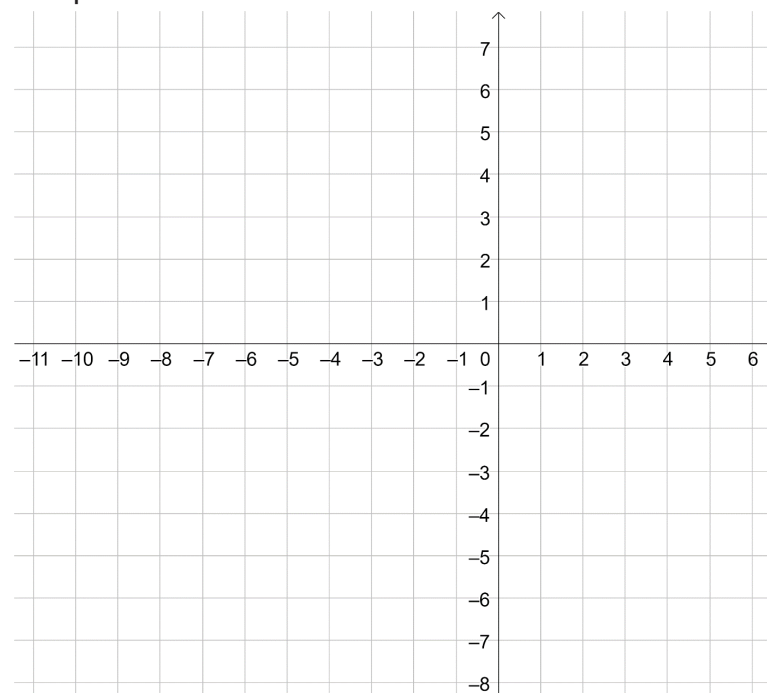
12.9.3 Your Turn!

1. The equation of circle R is $(x + 2)^2 + y^2 = 36$.

- a. Complete the table.

Center	
Radius	
Domain	$\underline{\hspace{1cm}} \leq x \leq \underline{\hspace{1cm}}$
Range	$\underline{\hspace{1cm}} \leq y \leq \underline{\hspace{1cm}}$

- b. Graph circle R .

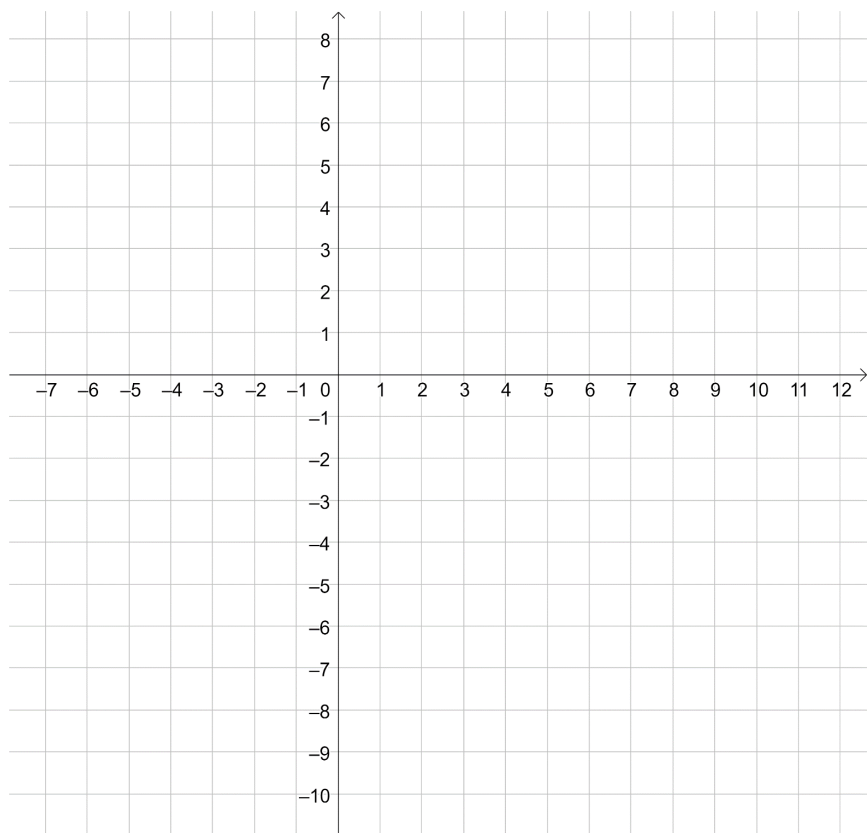


2. Alyona used the equation $(x - 3)^2 + (y + 1) = 81$ to determine how far she could sit, in meters, from her Xbox and still use her wireless controller.

a. Complete the table.

Center	
Radius	
Domain	$\underline{\hspace{1cm}} \leq x \leq \underline{\hspace{1cm}}$
Range	$\underline{\hspace{1cm}} \leq y \leq \underline{\hspace{1cm}}$

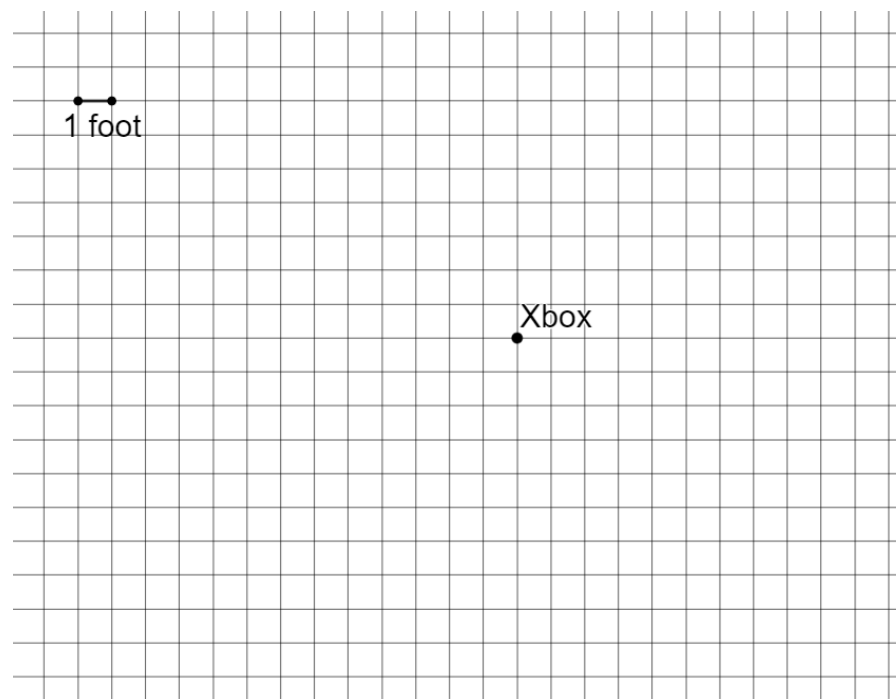
b. Graph $(x - 3)^2 + (y + 1) = 81$



c. What does the center represent in the context?

d. What does the radius represent in the context?

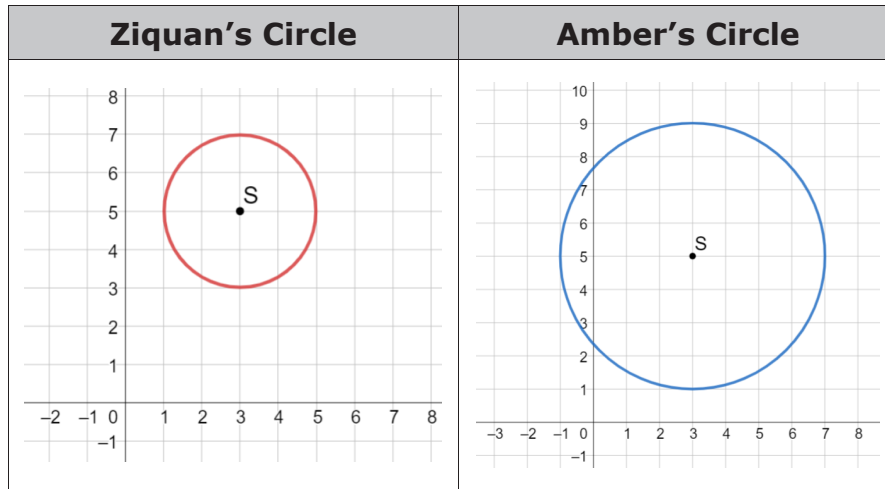
e. The grid shows the location of Alyona's Xbox. Use the domain and range to show all of the locations at which Alyona can sit.





12.9.4 Wrap-Up: Error Analysis

1. Ziquan and Amber both graphed circle S , $(x - 3)^2 + (y - 5)^2 = 4$, but their circles looked different.



- a. Who correctly graphed circle S ?
- b. Write a note to the person who graphed circle S incorrectly explaining their mistake.

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